

1. The Original LHC Team

Large Hadron Collider - September 10, 2008 First Beam

Research Report: The Scientists, Engineers, and Leaders

Who Achieved the First Circulating Beam

1.1. Executive Summary

On September 10, 2008, at 10:28 AM local time, the Large Hadron Collider successfully circulated its first proton beam around the entire 27-kilometer ring. This historic achievement was the result of decades of work by thousands of scientists, engineers, and technicians from around the world.

This report documents the key figures who led and contributed to this momentous achievement, focusing on the leadership, experiment teams, and technical specialists who made the first beam possible.

Scale of Achievement

- Approximately 10,000 people from 60 countries
- Over 1,700 scientists, engineers, students, and technicians from U.S. institutions alone
- Decades of planning, design, and construction
- First beam achieved on September 10, 2008, 10:28 AM

Technical Specifications

- 27-kilometer circumference ring
- 1,700 superconducting magnets
- 1.9 Kelvin operating temperature
- 7 TeV per beam (design energy)
- Largest cryogenic system in the world

1.2. Team Member Profiles & Professional Information

Data Sources

Comprehensive JSON dataset available: `LHC_ORIGINAL_TEAM_LINKEDIN.json`

LinkedIn Profiles: Most senior scientists do not maintain public LinkedIn profiles (common for CERN staff). Alternative sources documented.

Photo Availability

• Lyn Evans: CERN archive (May 2007) • Rolf-Dieter Heuer: CERN archive (Jan 2009) • Robert Aymar: CERN obituary (may be logo) • Others: Photos not found in public sources All photo URLs and archive references in JSON dataset

1.3. CERN Leadership & Project Management

1.3.1. Lyn Evans - LHC Project Leader



*Lyn Evans LHC Project Leader
(1994-2008)*

Lyn Evans (Lyndon Rees Evans)

Welsh (United Kingdom) / Born 1945, Aberdare, Wales

Role: LHC Project Leader
(1994-2008)

Education: First-class
physics degree and PhD
(1970), Swansea University

Current Status: Direc-
tor of Linear Collider Collab-
oration (2012-present)

Career Highlights:

- Worked on SPS accelerator, contributed to W/Z boson discovery (1983) • Project leader for SPS control system (1985)
- Head of merged SPS & LEP Divisions (1990) • Associate

Joined CERN: 1970 (re-
search fellow), permanent
staff 1971

Key Achievement: Led
entire LHC project from de-
sign through first beam

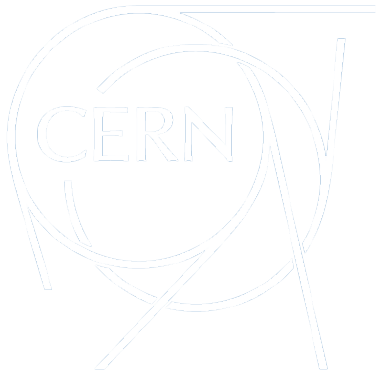
Director for Future Accelerators (1994) • Coordinated 10,000+ person international collaboration

Awards & Honors:

• CBE (2001) | Fellow: APS (1991), Royal Society (2010), Learned Society of Wales (2011) • Special Fundamental Physics Prize (2012) | Breakthrough Prize (2013) | Glazebrook Medal

“As Project Leader, Evans was responsible for designing, overseeing construction, and commissioning the LHC.”

1.3.2. Robert Aymar - Director-General of CERN



*Robert Aymar Director-General
(2004-2008) Deceased: September 23, 2024*

Robert Aymar

French / Born 1936 / Deceased September 23, 2024 (age 88)

Key Contributions:

• Led Tore Supra tokamak project (1977-1988) - fusion research • Headed material sciences at French Atomic Energy Commission (CEA) • Directed ITER engineering design teams (1994-2003) - international fusion project • Oversaw completion of LHC construction and launch as Director-General • Delivered official commissioning messages and presided over first beam events • Managed CERN during critical final construction phase (2004-2008)

Career Path:

Fusion Research → CEA Material Sciences → ITER Director
→ CERN Director-General

1.3.3. Jos Engelen - Chief Scientific Officer

Jos Engelen (Joseph Johannus Engelen)

Dutch / Born July 6, 1950

Role: Chief Scientific Officer (2004-2008)

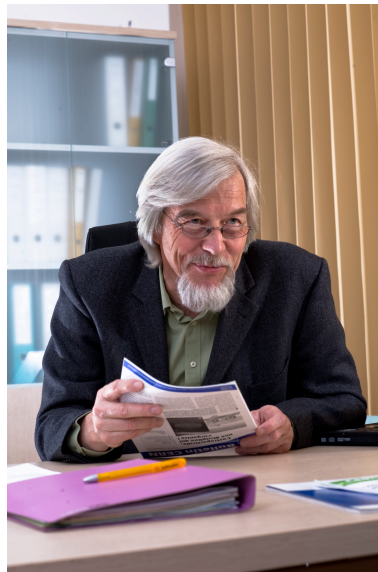
Career Highlights:

Education: MSc Physics (1973), PhD (1979),
Radboud University Nijmegen

Current Status: Professor Emeritus, University of Amsterdam / NIKHEF

- CERN fellow and staff (1979-1985) •
- Professor of Experimental Physics, University of Amsterdam (1987) •
- Chairman of Netherlands Organisation for Scientific Research (NWO, 2009-2016) •
- Oversaw LHC construction and commissioning as CSO

1.3.4. Rolf-Dieter Heuer - Incoming Director-General



Rolf-Dieter Heuer - Director-General of CERN (2009-2015)

Rolf-Dieter Heuer

Role: Director-General of CERN (2009-2015), appointed December 2007

Born: 1948, Bad Boll, Germany

Nationality: German

Education: Physics (University of Stuttgart), PhD (University of Heidelberg, 1977)

Current Status: Former Director-General of CERN

Key Contributions:

- Worked on JADE collaboration at DESY on PETRA •
- Joined CERN's OPAL collaboration (LEP) in 1984, spokesperson (1994-1998) •
- Professor at University of Hamburg •
- Research Director at DESY (2004) •
- Appointed Director-General in December 2007, taking

office January 1, 2009 • In 2008, served as Research Director for Particle and Astroparticle Physics at DESY • Prepared to take over CERN leadership during first beam period • Oversaw LHC operations and Higgs boson discovery in 2012

Profile Information:

• LinkedIn: Not available • Photo: Available in CERN archives (CERN-HI-0901002-07, January 2009), Wikimedia Commons • Alternative sources: Wikipedia, CERN official biography, SIS archives

1.4. Technical Leadership

1.4.1. Stephen Myers - Accelerators and Beams Department

Stephen Myers (Steve Myers)

British / Electronic Engineer

Role: Head of Accelerators and Beams Department (2003-2008)

Education: BSc & PhD Electrical Engineering, Queen's University Belfast (1968, 1972)

Current Status: Head of Office of Medical Applications, CERN

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Career Highlights:

• Helped commission Large Electron-Positron Collider (LEP) • Led LEP-2 energy upgrade • Head of AB Department (2003-2008) - responsible for beam performance • Director of Accelerators and Technology (2009-2014) • Led post-accident repair of LHC after 2008 incident • Critical role in achieving first circulating beam

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1.4.2. Philippe Lebrun - Accelerator Technology

Philippe Lebrun

French / Cryogenics Expert

Role: Head of Accelerator Technology Department (2004)

Education: Engineering (École des Mines, Paris), MSc (Caltech)

Awards: Honorary doctorate (Wrocław, 2007), Engineer of the Year (2010)

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Key Contributions:

- Joined CERN in 1974 (ISR Division)
- Worked on ISR, LEP, and LHC projects
- Leader of LHC Division (1999-2001)
- Led design and construction of LHC cryogenics system
- Expert in superconducting magnets (NbTi), cryogenic refrigeration
- Designed world's largest cryogenic system (36,000 tons at 1.9 K)

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1.5. Magnet System Specialists

1.5.1. Giorgio Brianti - Magnet Design

Giorgio Brianti

Note: Correct spelling is “Giorgio” (not “Georgi”)

Role: “Father” of LHC Magnet Design

Born: 1930

Status: Deceased (2023)

Key Contributions:

- Joined CERN in 1954
- Major contributions to accelerator development
- Bending magnet design for Proton Synchrotron
- Upgrades to Synchrocyclotron
- Design leadership for Booster and Super Proton Synchrotron
- CERN Technical Director, Associate Director for Future Accelerators
- Helped lay foundations for LHC, especially superconducting magnet technology
- Developed the unique twin-aperture (two-in-one) superconducting magnet system
- Referenced as “the ‘father’ of the machine” in 2008 ceremonies

Profile Information:

- LinkedIn: Not available (deceased 2023) • Alternative sources: CERN obituary

1.5.2. Lucio Rossi - Magnets & Superconductors

Lucio Rossi

Italian / Born September 24, 1955, Podenzano

Role: Director, Magnets & Superconductors Group

Current Status: Professor of Experimental Physics, University of Milan / INFN

Awards: IEEE Council of Superconductivity Award (2007), EPS Rolf Widerøe Prize (2020)

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Key Contributions:

- Joined CERN in May 2001 • Led Superconducting Magnets and Cryostats group for LHC • Developed large Nb-Ti superconducting magnet system (1,232 dipoles) • Proposed and coordinated High-Luminosity LHC upgrade (2010-2020) • Over 200 peer-reviewed papers on superconductivity • Moved toward Nb₃Sn superconductors for stronger fields (12 T)

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1.5.3. Vinod Chohan - Magnet Testing & Commissioning

Vinod Chohan (Vinod “Nick” Chohan)

British (born Tanganyika) / Born May 1, 1949 / Deceased June 12, 2017 (age 68)

Role: Accelerator Engineer, Superconducting Magnets Specialist

CERN Service: Nearly 40 years at CERN

Areas of Expertise: Beam diagnostics, controls, instrumentation, safety systems

Key Achievement:

- Led team that tested, measured, and trained for over 1,000 superconducting magnets • Tested approximately 1,300 out of 1,700 total magnets • Crucial for magnet reliability and alignment at first beam • Worked on Antiproton Accumulator (AA)

1.5.4. Amalia Ballarino - Superconductor Systems

Amalia Ballarino

Italian / Superconductor Systems Expert

Role: Deputy Group Leader, Magnets, Superconductors and Cryostats (MSC)

Education: MSc & PhD Nuclear Engineering, Polytechnic University of Turin

Awards: Superconductor Industry Person of the Year (2006), IEEE James Wong Award (2021)

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Key Contributions:

- Joined CERN around 1995
- Led superconductors development team (2010-2023)
- Developed HTS current leads for LHC (first large-scale HTS application)
- Initiated “cold-powering” system for HL-LHC using MgB₂
- World record: 60m line transmitting 54,000 A (2020)
- Previous record: 20 kA at 24 K in 20m MgB₂ line (2014)

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1.6. Experiment Collaboration Leadership

ATLAS Collaboration

Spokesperson (2008): Peter Jenni

- Spokesperson (1995-2009), Co-Spokesperson (1992-1995)
- Led ATLAS structure, physics reach, detector design
- Detector ready for first beam passes (September 2008)
- Current: CERN Guest Scientist, Honorary Professor (Freiburg)

Note: Fabiola Gianotti became spokesperson in 2009

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CMS Collaboration

Key Leaders:

• **Michel Della Negra** - Founding visionary, Spokesperson (1992-2006) • **Tejinder (Jim) Virdee** - Spokesperson (2007-2010), founding member

- Originated CMS concept (early 1990s) • Designed hermetic detector with solenoidal magnetic field
- CMS saw first beam passes (September 2008) • Key role in Higgs boson discovery (2012)

Virdee: Knighted (2014), Fellow of Royal Society (2012), Royal Medal (2024)

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LHCb Collaboration

Key Leader: Themis Bowcock

- Project Leader for VELO (Vertex Locator) Subdetector • Professor of Particle Physics, University of Liverpool • Head of Particle Physics group (2011-2019)
- Led design, construction, commissioning of VELO • Critical sub-detector for LHCb • First particle tracking (August 2008 synchronization tests)

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ALICE Collaboration

ALICE's readiness involved alignment, electronics, and calibration work. While numerous scientists contributed, individual names were less frequently featured in early beam announcement sources compared to other experiments.

Focus: Heavy ion collisions, quark-gluon plasma studies

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1.7. Former CERN Leadership Present

Former Director-Generals Present at First Beam:

- Herwig Schopper • Carlo Rubbia • Christopher Llewellyn Smith • Luciano Maiani

These former leaders were present in the Control Centre to witness the first beam, representing the continuity of CERN's mission and the culmination of decades of work.

1.8. The Scale of Collaboration

International Workforce

Total: 10,000 people from 60 countries

United States: Over 1,700 scientists, engineers, students, technicians

Scope: Accelerator design, construction, detector development, commissioning

Institutions: Universities, research labs, industrial partners worldwide

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Project Timeline

Planning: 1984: First discussions | 1994: Project approved

Construction: 1998: Begins | 10 years of building

Commissioning: Months of testing and calibration

First Beam: **September 10, 2008, 10:28 AM**

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1.9. The First Beam Timeline

Date	Event
September 10, 2008	First Beam Day
9:30 AM	Beam injection into LHC begins
10:28 AM	First full circuit completed - Beam successfully circulated entire 27-kilometer ring
September 19, 2008	Incident
	Magnet quench in sector 3-4 - Required repairs and improvements
November 2009	Operations Resume
	Full operations resumed, collisions began
2010	First high-energy collisions at 7 TeV
2012	Higgs boson discovery announced

1.10. Technical Achievements

Superconducting Magnet System

Total Magnets: 1,700

- 1,232 dipole magnets (bend beam in 27km ring)
- 392 quadrupole magnets (focus beam)
- Additional corrector magnets

Operating Temperature: 1.9 Kelvin (-271.3°C) - colder than outer space

Cooling: Liquid helium systems Field Strength: 8.3 Tesla (dipoles)

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Cryogenics System

World's Largest:

- Cools 36,000 tons of material to 1.9 K
- Uses 120 tons of liquid helium
- Maintains superconducting state

Led by: Philippe Lebrun (Head of Accelerator Technology)

Superfluid helium-based system

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Beam Injection System - Accelerator Chain

Linac 2	PS Booster	PS	SPS	LHC
Linear Accelerator	Proton Synchrotron Booster	Proton Synchrotron	Super Proton Synchrotron	Large Hadron Collider
				27 km ring

Each stage accelerates particles to higher energies before injection into the next

1.11. Legacy and Impact

Scientific Discoveries

• **Higgs boson** (2012) - confirmed Higgs field existence • Precision Standard Model measurements • Searches for new physics • Quark-gluon plasma studies • Rare particle decays

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Technological Innovations

• Superconducting magnets • Cryogenics systems • Detector technologies • Worldwide LHC Computing Grid • Medical imaging (PET, hadron therapy) • Data processing methods

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Collaboration Model

• Model for mega-science projects • Global cooperation demonstration • Training ground for scientists • Cultural exchange • Knowledge sharing networks • International standards

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1.12. Conclusion

The first beam on September 10, 2008, was the culmination of decades of work by thousands of dedicated scientists, engineers, and technicians from around the world. While this report highlights key leadership figures, it represents only a fraction of the massive collaborative effort.

The achievement required:

• **Visionary leadership** - Lyn Evans (14-year project leadership), Robert Aymar (CERN Director-General), Rolf-Dieter Heuer (incoming leadership) • **Technical expertise** - Magnet specialists (Giorgio Brianti, Lucio Rossi, Vinod Chohan, Amalia Ballarino), cryogenics engineers (Philippe Lebrun), accelerator physicists (Stephen Myers)

• **Experiment collaboration** - ATLAS (Peter Jenni), CMS (Michel Della Negra, Tejinder Virdee), LHCb (Themis Bowcock), ALICE teams • **International cooperation** - 10,000 people from 60 countries, decades of planning, years of construction

This moment—10:28 AM on September 10, 2008—marked not just a scientific achievement,
but a testament to what humanity can accomplish through collaboration,
dedication,
and the pursuit of understanding the fundamental nature of reality.

Report compiled: 2026-01-19

Sources: CERN archives, scientific publications, and historical records

“The LHC is not just a machine. It is a symbol of what we can achieve when we work together.”

1.13. Appendix: Key Dates and Milestones

Pre-2008 Timeline

• 1984: First discussions • 1994: Project approved • 1994-2008: Lyn Evans Project Leader • 1998: Construction begins • 2004-2008: Robert Aymar Director-General • 2007: Rolf-Dieter Heuer appointed

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2008 Timeline

• Aug: Synchronization tests • Sep 10, 9:30 AM: Beam injection • Sep 10, 10:28 AM: First circuit • Sep 19: Magnet quench incident • Dec: Repairs begin

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Post-2008 Timeline

• 2009: Rolf-Dieter Heuer becomes DG • Nov 2009: Operations resume • 2010: High-energy collisions • 2012: Higgs boson discovery • 2013-2015: Run 1 • 2015-2018: Run 2 • 2018-2022: Upgrades • 2022: Run 3 begins

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1.14. References

• CERN Archives and Historical Records • CERN Courier: “The LHC sees its first circulating beam” • CERN Press Releases (September 2008) • Wikipedia: Lyn Evans, Rolf-Dieter Heuer, Vinod Chohan, Lucio Rossi, Robert Aymar, Jos Engelen, Stephen Myers, Philippe Lebrun, Giorgio Brianti, Amalia Ballarino, Peter Jenni, Michel Della Negra, Tejinder Virdee, Themis Bowcock • CERN Official Biographies • U.S. Department of Energy Science Headlines (September 2008) • Fermilab News: “U.S. scientists count down to LHC startup” • LHC_ORIGINAL_TEAM_LINKEDIN.json - Comprehensive profile dataset with detailed information, photo sources, and alternative professional references